

Module 4 — Read 0

Project preview — Project 4 — Designing Fair Comparisons

Module 4, Read 0: Why fair comparisons are hard

Project: [Project 4, Designing Fair Comparisons](#) · Next: [Read 1](#)

The central question of Module 4

Almost every interesting claim in science and public policy involves comparing two groups:

- Does this drug work better than a placebo?
- Does the tutoring program raise test scores?
- Are workers in remote roles more productive than in-office workers?

These questions look straightforward. They are not. How you compare two groups determines whether your answer means anything.

What you will do in Project 4

Project 4 assigns you a research question about a two-group comparison. You will:

1. Classify the design, Is it an RCT or an observational study? Who decides group membership?
2. Identify confounders, Draw a simple DAG showing what else might explain the difference.
3. Simulate a dataset, Use `set.seed()` to create a reproducible, fictional dataset that mirrors the design.
4. Run the right test, Pick from the methods table by variable types (numerical → Welch t ; binary → two-proportion z).
5. Write for a classmate, Summarize your findings in one page for someone who asked the question but is not a statistician.

Your project stays limited to a single two-group comparison (one binary explanatory contrast). For a complete worked example, see the [Project 4 sample R notebook](#) (simulate → describe → test → check the confounder).

Three things that can go wrong when comparing groups

1. Confounding

A third variable affects both who ends up in each group and the outcome. Observational comparisons are vulnerable to confounding, even with large samples and small p-values.

2. Using the wrong test

Comparing two group *means*? Use the Welch two-sample t -test. Comparing two group *proportions*? Use a two-proportion z -test or chi-square. The [methods map](#) will guide you.

💡 3. Overclaiming causation

A p-value of 0.001 does not mean the treatment *caused* the difference in an observational study. Causal language requires a randomized experiment (or very careful design).

Module 4 learning path

Step	Material
Read 0 (<i>this page</i>)	Project preview, what makes comparisons fair
Read 1	Three sections: study design & confounding · two-group tests/CIs · more than two groups
Quiz A (Def & Notation)	Study-design vocabulary + two-group methods & output reading
Demo lab 1	Simulate data by study design; compare groups with Module 3 plots/stats
Demo lab 2	Inference code: t-test, prop.test/chisq.test, paired tests, ANOVA, glm
Exercise lab	Practice problems → connect to Project 4 plan
Synthesis quiz	Full integration: design + test + interpret
Milestone	Optional worksheet: draft design plan for your assigned question
Project 4	Final report + reproducible notebook

No lecture or lab numbers appear on the course path, just the content above.

Quick vocabulary preview

Term	Plain language
Explanatory variable (EV)	The group label, treatment, exposure, or condition
Response variable (RV)	The outcome you measure
RCT	Researcher flips a coin to assign groups, gold standard for causation
Observational study	Groups form on their own, researcher just watches
Confounder	A third variable that muddles the EV → RV comparison
DAG	A picture of causal paths between variables
Welch <i>t</i> -test	The R default for comparing two group means

i Before you begin Read 1

Think of a comparison you have seen in the news recently, a diet study, a drug trial, a social intervention. Ask yourself:

1. How did people end up in each group?
2. What else might differ between the groups?
3. Was the response variable numerical or binary?

You will have the vocabulary to answer these questions after Read 1.